

Problem-Solution Fit

Date	29 June 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being (HDI Prediction using AI/ML)
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] Policy researchers, government analysts, economists, NGO workers, students and academicians studying human development and country-level well-being indicators.	6. CUSTOMER LIMITATIONS [CL] Limited access to real-time HDI data. Manual calculation is time-consuming and error-prone. Non-technical users lack ML expertise to build prediction tools themselves.	5. AVAILABLE SOLUTIONS PROS & CONS [AS] PROS: UNDP HDI reports are authoritative. Excel tools allow manual calculation. CONS: Reports are annual and delayed. No interactive prediction tool. No instant country-level forecast for custom input values.
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] Difficulty predicting HDI scores for custom indicators without expert knowledge. Frequency: Researchers face this challenge every time new socioeconomic data is collected or policy scenarios are tested.	9. PROBLEM ROOT / CAUSE [RC] HDI data is multi-dimensional and complex. Existing tools are static and report-based. There is no accessible ML-powered tool that accepts custom inputs and instantly predicts HDI classification.	7. BEHAVIOR + ITS INTENSITY [BE] Users manually look up HDI tables, cross-reference multiple indicators, and compare country data. Intensity: HIGH — performed frequently by researchers and students during analysis and reporting cycles.
3. TRIGGERS TO ACT [TR] Need to evaluate HDI for a custom country scenario. Policy planning for development initiatives. Academic assignments on human development. Awareness of HDI disparities across nations.	10. YOUR SOLUTION [SL] A Flask-based ML web application that accepts Life Expectancy, Mean Years of Schooling, GNI per capita, and Internet Users as inputs and predicts the HDI score and category (Low / Medium / High / Very High) instantly using a trained Linear Regression model.	8. CHANNELS OF BEHAVIOR [CH] ONLINE Kaggle (dataset), GitHub (version control), SkillWallet (project tracking), Flask web app (browser-based prediction) OFFLINE Local machine development using VS Code, Jupyter Notebook, Anaconda
4. EMOTIONS BEFORE / AFTER [EM] BEFORE: Frustrated by manual HDI lookup, confused by complex		

tables, uncertain about predictions. AFTER: Confident and informed. Users get instant HDI predictions with clear classification, saving time and effort.		
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